

Amrutha Nagaraj

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https://ammun99.github.io/Personal_Portfolio/

OBJECTIVE

Full-Stack Software Engineer with experience designing scalable web applications, data pipelines, and research-focused platforms using modern JavaScript frameworks, Python, and cloud technologies. Interested in building secure scientific data systems, distributed research workflows, and user-centered computational tools.

EDUCATION

University of Central Florida

Orlando, FL

Masters in Computer Engineering

December 2024

- Software Development & Networks, Analysis of Algorithm, Machine Learning, Artificial Intelligence, Cybersecurity, Software Engineering, Computer Design & Computer Architecture, Computational Complexity

Visvesvaraya Technological University

Bengaluru, India

Bachelor in Engineering in Computer Science and Engineering

- Data Structures, Data Base Management System, Operating Systems, Big Data Analytics, Object Oriented Programming, Computer Organization and Architecture, Automata Theory and Computability

EXPERIENCE

Software Engineer (Full Stack)

March 2025 - Present

KGS Technology Financial – React, JavaScript, TypeScript, Python, Node.js, AWS

Alpharetta, GA

- Designed and delivered end-to-end full stack features using React, Python, and Node.js, improving usability and workflow efficiency across internal financial platforms.
- Built responsive, accessible web interfaces ensuring cross-browser compatibility and seamless integration with RESTful backend APIs.
- Built responsive and accessible UI components following modern design patterns, optimizing layout behavior across devices and improving usability of complex data-driven interfaces.
- Developed and integrated secure REST APIs, enabling real-time validation, data synchronization, and reliable user interactions.
- Optimized database queries and backend processing to enhance performance of high-volume transactional systems and reduce latency.
- Collaborated in Agile teams and monitored production deployments using AWS CloudWatch to improve reliability and release quality.

Graduate Teaching Assistant

2022 - 2024

University of Central Florida – Java, JavaScript, React, Python, Azure

Orlando, FL

- Mentored 200+ students in full stack web development, covering frontend, REST APIs, Vue, Tailwind, Python, React and JavaScript, authentication, and database integration.
- Guided teams in building responsive and user-centered applications emphasizing accessibility, UI usability, and modern CSS layout techniques.
- Led labs on containerized deployments using Docker and CI/CD workflows for scalable application delivery.
- Evaluated Human-Computer Interaction projects focused on interface clarity, interaction flow, and mobile-first responsive design strategies.
- Documented architecture solutions and debugging practices to support knowledge transfer and maintainable software design.

Front-end Web Developer

July 2021 - November 2021

Learnovate E-Commerce – React.js, Vue.js, GraphQL, JavaScript, AWS, SQL Server

Bengaluru, India

- Developed responsive Single Page Application features using React and Vue.js, improving interaction flow and frontend performance.
- Integrated UI components with backend services via REST and GraphQL APIs, enabling real-time rendering.
- Contributed to performance optimization and cloud deployments on AWS Fargate ECS while collaborating with backend teams for end-to-end delivery.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C, C++, Go

Frontend: React, Vue.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, Django, REST APIs, Authentication (JWT/OAuth)

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Redis

Cloud & DevOps: AWS (EC2, RDS, Lambda, CloudWatch), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins)

Data & Streaming: Apache Kafka, Spark Streaming, ETL Pipelines

PROJECTS

- Flying Bird Game** | *Java, Vue 3, Vite, OOP, Game Loop Architecture* 2025
- Built a Bird-inspired 2D arcade game in Java featuring physics-based bird motion, real-time collision detection, and procedural obstacle generation using Swing graphics.
 - Implemented event-driven gameplay architecture with a structured game loop, separating rendering, physics, and input handling into maintainable OOP components.
 - Deployed the game as an interactive Vue 3 web page, enabling browser-based launching with a live demo at ammun99.github.io/bird-game-page.
- Personal Portfolio Website** | *Vue 3, JavaScript, Tailwind CSS, GitHub* 2025
- Architected a production-grade, component-driven frontend using Vue 3, focusing on reusable UI patterns and responsive page composition.
 - Designed and implemented a responsive, mobile-first interface using Tailwind CSS, optimizing visual hierarchy, usability, and interaction flow.
 - Engineered performance-conscious UI interactions with GSAP, improving responsiveness, navigation flow, and presentation quality.
- ETL-Driven SEO Analytics Platform** | *Python, React, Node.js, Express.js, MySQL, REST APIs* 2025
- Designed ETL pipelines extracting data from the Google Search Console API, transforming datasets, and loading structured data into MySQL.
 - Developed backend APIs using Node.js and Express to serve analytics insights to a responsive React dashboard.
 - Optimized API performance & frontend data rendering workflows to support interactive exploration and reporting.
- Library Management System** | *TypeScript, React, Redux, Redis, Django, PostgreSQL, GraphQL, Unit Testing* 2024
- Designed frontend with React and TypeScript, enabling administrators to manage user operations effortlessly, designed with accessibility and responsiveness in mind to serve a wide range of users efficiently.
 - Contributed iterative improvements by implementing APIs using Django & PostgreSQL, GraphQL for robust data storage, enhancing the efficiency and security of user management functionality.
- Real-Time Analytics Pipeline** | *Kafka, Spark Streaming, Python, Docker, SQL* 2024
- Designed distributed streaming pipelines using Apache Kafka and Spark Streaming for real-time ingestion and transformation of large datasets.
 - Implemented aggregation workflows and fault-tolerant processing logic to support scalable analytics delivery.
 - Containerized data processing components using Docker, enabling reproducible deployment across environments.
- Research Data Processing Platform** | *Java, Spring Boot, PostgreSQL, AWS, Microservices* 2023
- Developed backend services using Java and Spring Boot to support secure data processing workflows and authenticated user access.
 - Designed PostgreSQL schemas using Spring Data JPA and JDBC, improving CRUD performance and reliability for structured data operations.
 - Deployed distributed backend components on AWS, following microservice architecture patterns for scalability and maintainability.